Sparse & Explainable Bayesian Neural Networks for Official Statistics

Unit-Weighted Pseudo-Posteriors, Formal Differential Privacy, and Non-Probability Propensity Estimation in LBBNN

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Section 1

PART I: Foundations of LBBNN & Input-Skip Architecture (ISLaB)

Motivation — What Official Statistics Truly Requires

The Primary Mandate of Official Statistics:

- **Unbiased Finite Population & Domain Inference**: Calibrated, design-consistent estimation of finite-population parameters (means, totals, sub-domain distributions) with valid uncertainty quantification.
- **Dual Transparency (Global & Local)**: Structural feature selection at the population level (global sparsity) and verifiable respondent-level sensitivity attributions (local polyhedral GLMs).

The Reality of Complex Modern Data Regimes:

- **Non-Linear Participation Bias**: In opt-in and administrative sources (S_c), self-selection mechanisms are non-linear functions of respondent covariates—standard linear logit models are structurally misspecified.
- **Differential Stratification & Informative Designs**: Sample inclusion probabilities differ across demographic strata, requiring **pseudo-posterior likelihood reweighting** to correct selection bias.
- **Formal Differential Privacy (ϵ, δ)**: Skewed economic microdata (wages, firm revenues) contain extreme leverage records that threaten re-identification, requiring privacy calibration that preserves distributional shape.
- **Weight-Variance Protection**: Dense deep models retain pure noise covariates, triggering severe variance inflation in downstream Horvitz–Thompson and Hájek weighting.

From Dense BNNs to Latent Binary BNNs (Hubin & Storvik, 2024)

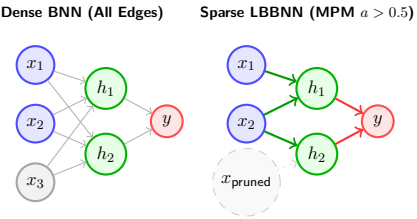
Spike-and-Slab Edge Selection:

- Latent binary inclusion indicator $\gamma_{k,p}^{(j)} \in \{0, 1\}$:

$$p\big(w_{k,p}^{(j)} \mid \gamma_{k,p}^{(j)}\big) = \gamma_{k,p}^{(j)} \mathcal{N}\big(0, (\tau_{k,p}^{(j)})^2\big) + \big(1 - \gamma_{k,p}^{(j)}\big)\delta_0$$

$$p\big(\gamma_{k,p}^{(j)}\big) = \text{Bernoulli}\big(\psi_{k,p}^{(j)}\big)$$

- **Bayesian Model Averaging (BMA)**: Ensembles across posterior draws $\Gamma \sim q(\Gamma)$ and $W \sim q(W \mid \Gamma)$ for optimal uncertainty quantification.
- **Median Probability Model (MPM)**: Hard-prunes edges with posterior inclusion probability $a_{k,p}^{(j)} \leq 0.5$ (Barbieri & Berger, 2004 optimal predictive submodel).



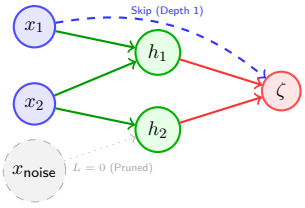
Input-Skip LBBNNs (ISLaB) & Structural Depth Metrics (Høyheim et al., 2026)

The ISLaB Architecture (*JAIR*, 2026):

- Direct skips route inputs $\mathbf{x} \in \mathbb{R}^p$ to every hidden layer: $\mathbf{a}^{(j-1)} = [a_1^{(j-1)}, \ldots, a_K^{(j-1)}, \mathbf{x}^\top]^\top$.
- **Active Path (AP)**: Connects input x_k to output ζ via edges with $a_{k,p}^{(j)} > 0.5$.

Contribution Depth Classification:

- **Depth = 1**: Direct skip $\implies$ linear GLM effect.
- **Depth $\geq$ 2**: Hidden layers $\implies$ non-linear interaction.
- **Paths = 0**: Noise covariate pruned from graph.



Local Explainability via Polyhedral ReLU Activation Regions

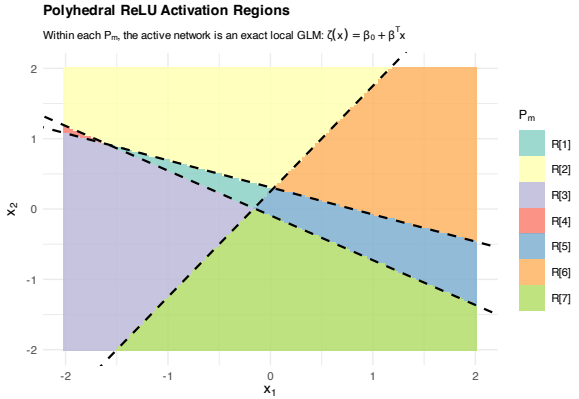
Local GLM Theorem (JSM 2026):

- Piecewise-linear activations partition $\mathbb{R}^p$ into polyhedral activation regions $\mathcal{P}_1, \ldots, \mathcal{P}_M$.
- Within each local region $\mathcal{P}_m$, the active network reduces to an exact **local GLM**:

$$\zeta(\mathbf{x}) = \beta_0(\mathbf{x}_i) + \sum_{j=1}^{p} \beta_j(\mathbf{x}_i) x_j$$

$$\boldsymbol{\beta}(\mathbf{x}_i) = \nabla_{\mathbf{x}} \zeta(\mathbf{x})\big|_{\mathbf{x}=\mathbf{x}_i}$$

- **Exact Gradients**: Evaluated via analytical backpropagation (zero perturbation error).
- **Credible Intervals**: Draws $W \sim q(W \mid \Gamma^*)$ yield exact posterior intervals for respondent sensitivities.



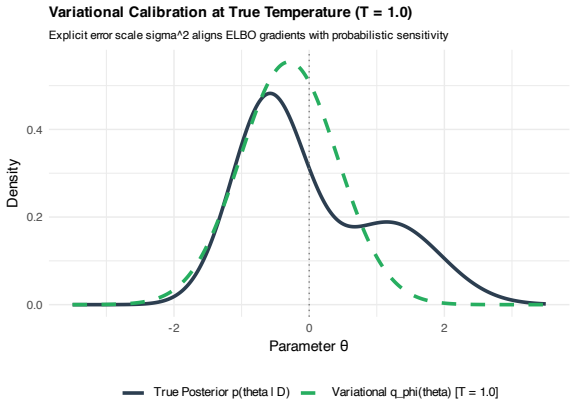
Variational Inference & Protection Against Cold Posteriors

Variational ELBO at True $T = 1.0$:

$$\text{ELBO}(q) = \sum_{i=1}^{n} \mathbb{E}_q[\log f(y_i \mid \mathbf{x}_i, \theta)] - \text{KL}(q \parallel p)$$

Why Explicit Error Scale σ^2 Matters (Wenzel et al., 2020):

- Omitting residual noise scale σ^2 artificially rescales likelihood ($T = \sigma^2 \neq 1$), mis-calibrating spike-and-slab KL shrinkage.
- In `lbbnn_net(sigma = ...)`, calibrated σ^2 grounds log-likelihood sensitivities on the true Bayesian scale.
- Calibrates both feature inclusion probabilities a_k and DP sensitivities f_i.



The LBBNN Open-Source Software Ecosystem

R Package LBBNN (CRAN v0.1.6):

- Pure C++ LibTorch backend via `torch` for R (zero Python dependencies or overhead).
- Mean-field spike-and-slab with Local Reparameterization Trick (LRT) and Normalizing Flows (*TMLR*, 2024).
- Automated model selection, predictive ensembling (BMA), and sparse median probability models (MPM).

Joint BLS Collaboration Modules:

- **Unit Weighting**: `get_dataloaders()` supports non-trainable weights $u_i \geq 0$ with optional sample-size normalization.
- **Residual Calibration**: `lbbnn_net(sigma = ...)` incorporates observation noise scale.
- **Differential Privacy**: `loglik_contributions()` and `fit_dp_lbbnn()` for multi-pass (ϵ, δ)-PDP synthesis.
- **Finite-Sample UQ & DP**: `adjust_slab_sandwich_lbbnn()` (Cholesky sandwich UQ) and `calibrate_dp_finite_sample()` ((ϵ, δ)-DP nugget calibration).
- **Composite Propensities**: Native losses for `STACK`, `LINK1`, `LINK2`, and `MIP`.

Section 2

PART II: Track 1 — Unit-Weighted Pseudo-Posteriors & Formal Differential Privacy

Unit-Weighted Variational Pseudo-Posteriors (Savitsky & Toth, 2016)

Equipping each unit $i \in \{1, \ldots, n\}$ with a non-negative weight $u_i \geq 0$ defines the pseudo-posterior:

$$p_{\mathbf{u}}(\theta \mid \mathbf{y}) \propto p(\theta) \prod_{i=1}^{n} f(y_i \mid \mathbf{x}_i, \theta)^{u_i}$$

$$\mathsf{ELBO}(q; \mathbf{u}) = \sum_{i=1}^{n} u_i \, \mathbb{E}_{q(\theta)}[\log f(y_i \mid \mathbf{x}_i, \theta)] - \mathsf{KL}\big(q(\theta) \parallel p(\theta)\big)$$

1. Survey Sampling Weights ($u_i = \tilde{w}_i$):

- Normalized to sample size n: $\tilde{w}_i = w_i \frac{n}{\sum w_j}$.
- Setting `normalize_weights = TRUE` ensures $\sum \tilde{w}_i = n$ does not overpower spike-and-slab KL shrinkage.
- Corrects informative sampling selection bias.

2. Differential Privacy Weights ($u_i = \alpha_i$):

- Unnormalized weights: $\alpha_i \in [0, 1]$.
- Setting `normalize_weights = FALSE` ensures $\sum \alpha_i \leq n$.
- Downweights high-sensitivity tail outliers and contracts the pseudo-posterior toward the prior.

Track 1A — Informative Survey Sampling Setup & Automated Noise Pruning

Simulation Setup ($N = 2,000, p = 4$):

- **True Non-Linear DGP**:

  $$Y_i = 2X_{1i} + 1.5\sin(\pi X_{2i}) + 0.8X_{1i}X_{2i} + \varepsilon_i$$

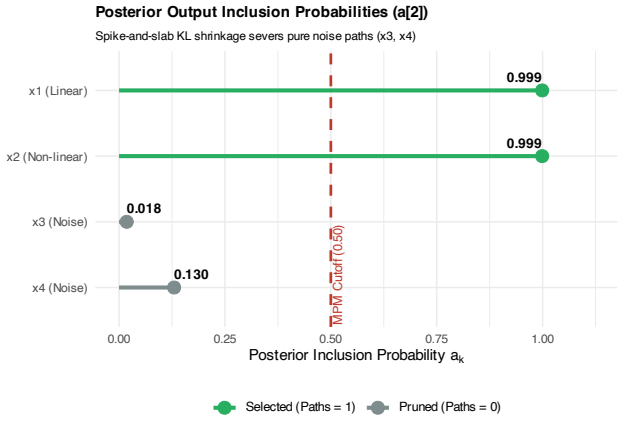
  where $\varepsilon_i \sim \mathcal{N}(0, 0.4^2)$, and $X_3, X_4 \sim \mathcal{N}(0,1)$ are pure noise.

- **Informative Selection Mechanism**:

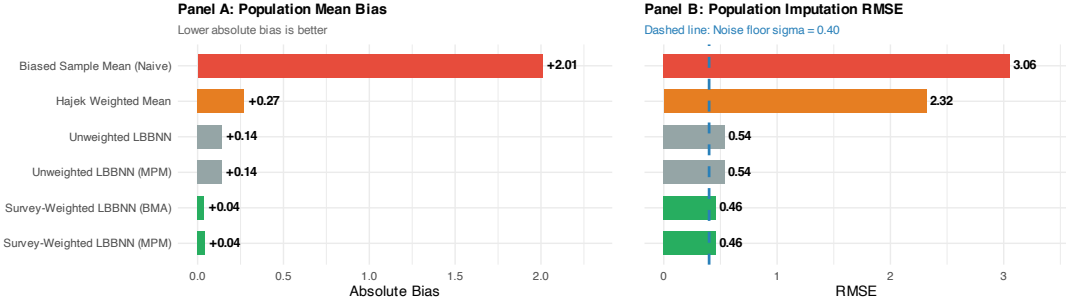
  $$\pi_i = \sigma(-1.0 + 1.2Y_i), \qquad \sigma(z) = \frac{1}{1+e^{-z}}$$

  Draws $n = 737$ units favoring high Y_i.

- Unweighted sample mean is **+2.009** (True population mean: **0.000**).



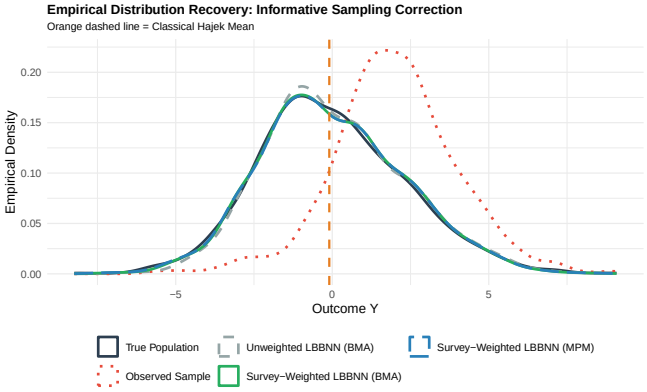
Track 1A — Population Mean & Imputation RMSE Recovery (Visual Benchmark)



Key Takeaways:

- Classical **Hájek mean** reduces bias ($+0.27$) but only provides a single scalar estimate (Pop RMSE $= 2.32$).
- **Survey-Weighted LBBNN** learns the conditional surface, reducing Pop RMSE to **0.46** (approaching $\sigma = 0.40$ noise floor).

Track 1A — Full Population Distribution Recovery



Distributional Recovery: Survey-weighted LBBNN models (green/blue curves) accurately reconstruct the full population density (black solid curve) across all 2,000 units, completely correcting the severe rightward shift of the observed sample (red dotted curve).

Track 1A — Domain Estimation, Imputation & Dirichlet Bootstrap UQ

Domain Totals & Conditional Expectations:

- **Domain Membership Matrix** $\mathbf{G} \in \mathbb{R}^{n \times D_{\text{dom}}}$ defines sub-populations:

$$\hat{\bar{Y}}_d^{\text{BNN}(s)} = \frac{\sum_{i=1}^{n} w_i G_{i,d}\, \hat{\mu}_i^{(s)}}{\sum_{i=1}^{n} w_i G_{i,d}}$$

- **Model-Assisted Conditional Mean**: Evaluating $\hat{\mu}_i^{(s)} = \mathbb{E}[Y_i \mid \mathbf{x}_i, \theta^{(s)}]$ (`type = "conditional_mean"`) estimates the smooth regression surface without adding simulation noise $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.
- **Item Non-Response Imputation**: Missing outcomes ($y_i = \text{NA}$) are filled via full posterior predictive draws $\tilde{y}_i^{(s)} \sim \mathcal{N}(\mu_{\theta^{(s)}}(\mathbf{x}_i), \sigma^2)$ (`impute_response_lbbnn`).

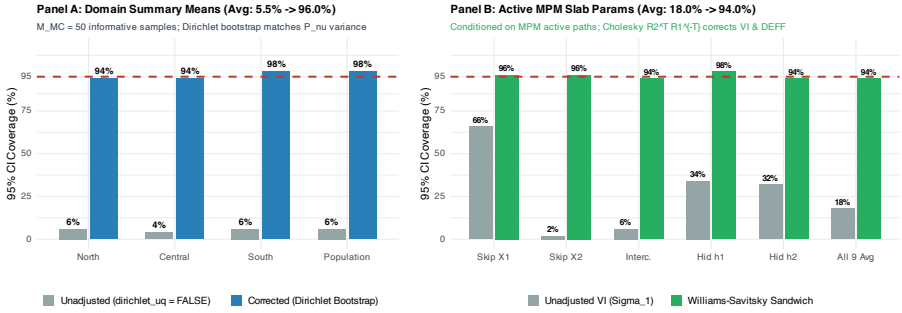
Dirichlet Bayesian Bootstrap UQ:

- **Pseudo-Posterior Variance Deficit**: Conditioning on fixed sample S captures parameter uncertainty $\theta^{(s)} \sim q(\theta)$ but underestimates design-based sampling variance.
- **Joint Parameter + Design Perturbations** (Savitsky & Toth, 2016): For each posterior draw $s \in \{1, \ldots, S\}$, sample $\theta^{(s)} \sim q(\theta)$ *and* flat weights $\boldsymbol{\xi}^{(s)} \sim \text{Dirichlet}(1, \ldots, 1)$:

$$\hat{\bar{Y}}_d^{(s)} = \frac{\sum_{i=1}^{n} \xi_i^{(s)} w_i G_{i,d}\, \hat{\mu}_i^{(s)}}{\sum_{i=1}^{n} \xi_i^{(s)} w_i G_{i,d}}$$

- Compounds epistemic BNN uncertainty with survey design variability (`dirichlet_uq = TRUE`).

Track 1A — Frequentist UQ Under P_ν: Sandwich vs. Dirichlet Bootstrap



Target Class A: Domain Summary Estimators

- Dirichlet Bayesian Bootstrap (`dirichlet_uq = TRUE`) restores 95% repeated-sampling coverage: **5.5%** $\rightarrow$ **96.0%** ($M_{\text{MC}} = 50$, Kish DEFF $= 2.25$).
- Compounds variational $\hat{\mu}_i^{(s)}$ with $\boldsymbol{\xi}^{(s)} \sim \text{Dirichlet}(1, \ldots, 1)$.

Target Class B: Active MPM Slab (`adjust_slab_sandwich_lbbnn`)

- Conditioned on MPM active paths (where BvM holds), Cholesky projection $R_2^\top R_1^{-\top}$ corrects diagonal VI + DEFF: **18.0%** $\rightarrow$ **94.0%** (`Skip_x1`: **66%** $\rightarrow$ **96%**).
- For hierarchical unit truth θ_i, sandwich extends to random-effect scores.

Pipeline & Return Structures: Survey-Weighted LBBNN (Track 1A)

```r
# 1. Package design weights & allow missing response values (y_i = NA)
dl_train <- get_dataloaders(df_survey, weights = w_design, normalize_weights = TRUE,
                            allow_missing = TRUE, train_batch_size = 64, test_batch_size = 32)
# 2. Build LBBNN with calibrated residual noise scale sigma (protects T=1)
net <- lbbnn_net(problem_type = "regression", sizes = c(p, 16, 1),
                 sigma = 0.4, input_skip = TRUE)
fit <- train_lbbnn(LBBNN = net, train_dl = dl_train$train_loader, epochs = 100, lr = 0.01)
# 3. Domain predictions & Bayesian imputation with Dirichlet uncertainty draws
dom_est <- predict_domains_lbbnn(net, X, domain_matrix = G, weights = w_design,
                                 type = "conditional_mean", dirichlet_uq = TRUE, draws = 50)
imp_res <- impute_response_lbbnn(net, newdata = df_survey, target_col = "y", draws = 50)
# 4. Williams-Savitsky (2021) Cholesky sandwich UQ on active MPM slab weights
slab_uq <- adjust_slab_sandwich_lbbnn(net, X, y = df_survey$y, weights = w_design, sigma = 0.4)
```

Step / Object	R Class / Type	Dimensions	Statistical Contents & Role
`dl_train$train_loader`	`torch::dataloader`	Batches: $B = 64$	Tensors `x` $[B, p]$, `y` $[B, 1]$, and weights `w` $[B, 1]$ ($\sum \tilde{w}_i = n$).
`net`	`lbbnn_net` (R6)	$p \to 16 \to 1$	Variational weights μ_W, σ_W, logits ψ; direct skips $\mathbf{x} \to h_j, \zeta$.
`dom_est`	`data.frame`	$D_{\text{dom}} \times 10$	Domain means, totals, posterior SDs, and 95% Dirichlet credible bounds.
`imp_res$completed_data`	`data.frame`	$n \times (p + 1)$	Completed dataset with missing y_i replaced by posterior mean $\hat{\mu}_i$.
`slab_uq$summary_table`	`data.frame`	$K_{\text{act}} \times 8$	MPM slab means, `Hessian_SD`, `Sandwich_SD`, `Survey_DEFF`, 95% CIs.

- **Survey Normalization**: `normalize_weights = TRUE` enforces $\sum \tilde{w}_i = n$ (Savitsky & Toth, 2016).

Track 1B — Why Unweighted Neural Networks Fail Differential Privacy

Probabilistic Differential Privacy (Savitsky, Williams & Hu, 2022):

$$\sup_{\theta \in \Theta_\delta} \left| \log \frac{q_{\boldsymbol{\alpha}}(\theta \mid \mathbf{y})}{q_{\boldsymbol{\alpha}}(\theta \mid \mathbf{y}')} \right| \leq \epsilon_{\mathbf{y}} = 2\Delta_{\alpha,\mathbf{y}}, \qquad \Delta_{\alpha,\mathbf{y}} = \max_{s,i} |\alpha_i \log f(y_i \mid \mathbf{x}_i, \theta^{(s)})|$$

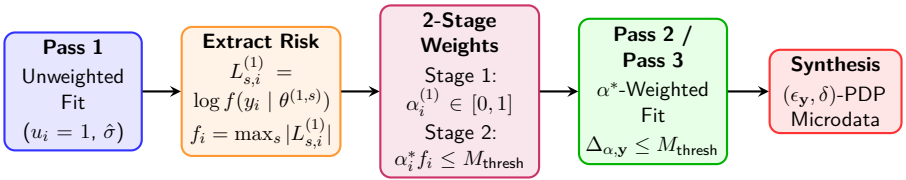
The Quadratic Exposure Dilemma in Gaussian Regression:

- Pointwise Gaussian log-likelihood grows quadratically with the residual:

$$|\log f(y_i \mid \mathbf{x}_i, \theta)| \propto \frac{(y_i - \mu_\theta(\mathbf{x}_i))^2}{2\sigma^2}$$

- In skewed survey microdata (wages, financial expenditures), extreme outliers drive unweighted sensitivity $\Delta \to \infty$ and privacy budget $\epsilon_{\mathbf{y}} \to \infty$.
- Naive gradient clipping distorts the mean and destroys the variance structure of official statistics.

Track 1B — Multi-Pass Privacy Calibration Pipeline (`fit_dp_lbbnn`)



Definitions & End-to-End Privacy Verification:

- **Pass 1 Pointwise Log-Likelihood ($L_{s,i}^{(1)}$)**: Evaluated across $s = 1, \ldots, S$ unweighted posterior draws $\theta^{(1,s)} \sim q^{(1)}(\theta)$ to measure worst-case record sensitivity $f_i = \max_s |L_{s,i}^{(1)}|$.
- **Two-Stage Weight Calibration ($\alpha_i^{(1)} \to \alpha_i^*$)**: Stage 1 maps f_i to an initial weight $\alpha_i^{(1)} \in [0,1]$; Stage 2 enforces the sensitivity threshold $\alpha_i^* f_i \le M_{\text{thresh}}$.
- **Empirical Lipschitz Bound**: After fitting the α^*-weighted pseudo-posterior (Pass 2, plus Pass 3 if vector rescaling shifts geometry), verify $\Delta_{\alpha,\mathbf{y}} = \max_{s,i} |\alpha_i^* L_{s,i}^{(\text{final})}| \le M_{\text{thresh}}$, guaranteeing $\epsilon_{\mathbf{y}} = 2\Delta_{\alpha,\mathbf{y}}$.

Track 1B — Stage 1: Initial Privacy Weight Normalization on f_i (Savitsky, 2026)

Direct Normalization of $f_i = \max_s |L_{s,i}^{(1)}|$ into $\alpha_i^{(1)}$:

1. **Linear Min-Max on f_i (`"linear"`):**

$$\tilde{f}_i = \frac{f_i - f_{\min}}{f_{\max} - f_{\min}}, \quad \alpha_i^{\text{lin}} = c(1-\tilde{f}_i) = c\frac{f_{\max} - f_i}{f_{\max} - f_{\min}}$$

2. **Percentile Deficit Truncation (`"percentile"`):**

$$\alpha_i^{\text{pct}} = \min\left(1,\ \max\left(0,\ \frac{q_{p_R}(f) - f_i}{q_{p_R}(f) - q_{p_L}(f)}\right)\right)$$

3. **Target Ratio Clipping (`"target_clip"`):**

$$\alpha_i^{\text{clip}} = \min\left(1,\ \frac{\Delta_{i,\text{goal}}}{f_i}\right)$$

Key Theoretical Insights (Savitsky, 2026):

- **Direct $1 - \tilde{f}_i$ vs. Reciprocal $1/f_i$ (Remark 1)**: Normalizing $1/f_i$ induces hyperbolic attenuation $\alpha_i^{\text{recip}} = (f_{\min}/f_i)\alpha_i^{\text{lin}}$, collapsing weights when $f_{\min} \to 0$. Direct $1 - \tilde{f}_i$ prevents collapse; $\alpha_i^{\text{lin}} f_i$ peaks at $f_{\max}/2$ ($\leq \frac{c}{4} f_{\max}$).
- **Percentile Boosting**: Sets $\alpha_i^{\text{pct}} = 1$ for $f_i \leq q_{p_L}(f)$ and 0 for $f_i \geq q_{p_R}(f)$, boosting $\sum \alpha_i^{(1)}$ while Stage 2 caps $\alpha_i^* f_i$.
- **Exact Target Control (Prop. 1)**: $\alpha_i^{\text{clip}} f_i \leq \Delta_{\text{goal}}$ by construction; all moderate units ($f_i \leq \Delta_{\text{goal}}$) retain full weight $\alpha_i^{(1)} = 1$.

Track 1B — Stage 2: Enforcing the Threshold ($\alpha_i^* f_i \le M_{\text{thresh}}$)

Second-Stage Reweighting Schemes ($\alpha_i^{(1)} \to \alpha_i^*$):

1. **Hard Truncation (`"hard_trunc"`)**:

$$\alpha_i^{\text{trunc}} = \alpha_i^{(1)} \, \mathbb{I}\Big(\alpha_i^{(1)} f_i \le M_{\text{thresh}}\Big)$$

Post-hoc zeroing of records where $\alpha_i^{(1)} f_i > M_{\text{thresh}}$; leaves unspent privacy capacity $M_{\text{thresh}} - \alpha_i^{(1)} f_i > 0$ on surviving units.

2. **Vector Upward Rescaling (`"vector_upscale"`)**:

$$\alpha_i^{\text{vec}} = \min\left(1,\, \alpha_i^{(1)} \cdot \frac{M_{\text{thresh}}}{\alpha_i^{(1)} f_i}\right) = \min\left(1,\, \frac{M_{\text{thresh}}}{f_i}\right)$$

Inflates all active weights ($\alpha_i^{(1)} > 0$) up to the privacy threshold boundary; pointwise dominates hard truncation ($\alpha_i^{\text{vec}} \ge \alpha_i^{\text{trunc}}$).

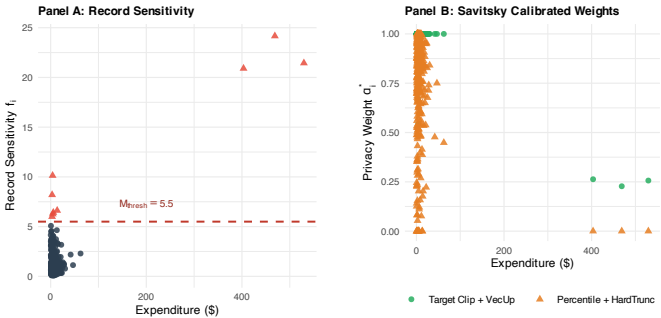
Information Dominance & Multi-Pass Geometry:

- **Algebraic Cancellation Insight**: For any record with $\alpha_i^{(1)} > 0$, $\alpha_i^{(1)}$ cancels in α_i^{vec}, collapsing directly to $\min(1, M_{\text{thresh}}/f_i)$ (`target_clip`). If Stage 1 zeroes tail units (`percentile`), those zeroes remain preserved.
- **Loewner Fisher Dominance (Prop. 2)**: Because $\alpha_i^{\text{vec}} \ge \alpha_i^{\text{trunc}}$ everywhere, vector upward rescaling strictly maximizes Fisher information:

$$\mathcal{I}_{\boldsymbol{\alpha}^{\text{vec}}}(\theta) \succeq \mathcal{I}_{\boldsymbol{\alpha}^{\text{trunc}}}(\theta)$$

- **Why Pass 3 Is Executed**: Upward rescaling shifts the pseudo-posterior geometry $\xi_{\boldsymbol{\alpha}^{\text{vec}}}(\theta \mid D)$; `fit_dp_lbbnn()` executes Pass 3 on α_i^{vec} to verify $\Delta_{\alpha,\mathbf{y}} \le M_{\text{thresh}}$ on the final variational draws.

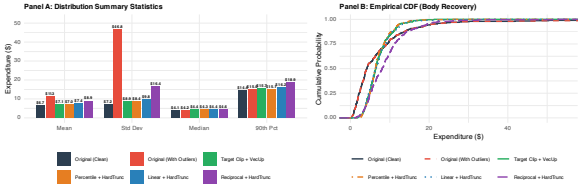
Track 1B — Record Sensitivity & Savitsky Two-Stage Weight Diagnostics



Empirical DP Calibration ($n = 300, M_{\text{thresh}} = 5.5$):

- **Skewed Microdata**: Median $\$4.16$, contaminated with 3 extreme outliers ($\$400$–$\$530$, $f_i \in [20.9, 24.2]$).
- **Likelihood Mass ESS**: In DP pseudo-posteriors, $\alpha_i^* \in [0, 1]$ acts as a likelihood exponent, so total retained information mass is ESS $= \sum_{i=1}^{n} \alpha_i^*$.
- **`target_clip + vector_upscale`**: Retains $\alpha_i^* = 1.0$ on 97.0% of sample (ESS $= 296.4/300$, 0 zeroed); caps outliers at $\alpha_i^* \in [0.19, 0.25]$.
- **`percentile + hard_trunc`**: Boosts lower 95% while hard-zeroing the top 5% tail (15 units zeroed, ESS $= 212.3/300$, $\epsilon_{\mathbf{y}} = 2.10$).

Track 1B — Full Factorial Benchmark Across Privacy Weight Schemes

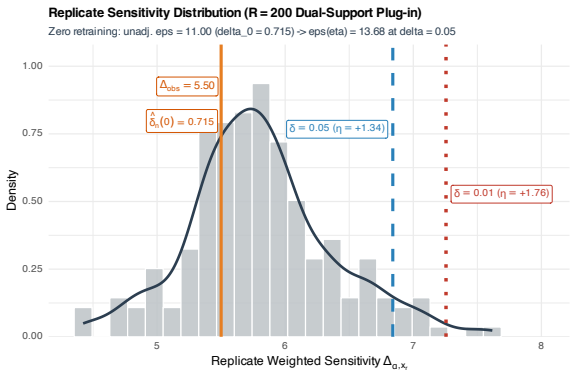


Full 3×2 Factorial + Reciprocal ($n = 300, M_{\text{thresh}} = 5.5$):

Stage 1	Stage 2	ESS	0s	$\epsilon_{\mathbf{y}}$	Mean (SD)
Clean Ref	—	300.0	0	—	\$6.70 (\$7.23)
Contam. ($u_i = 1$)	—	300.0	0	48.33	\$11.30 (\$46.75)
`target_clip`	`vec_up`	**296.4**	**0**	11.00	**\$7.13** (**\$8.92**)
`target_clip`	`hard_tr`	296.4	0	11.00	\$7.13 (\$8.92)
`linear`	`vec_up`	296.2	1	11.00	\$7.13 (\$8.92)
`linear`	`hard_tr`	239.3	1	10.03	\$7.42 (\$9.85)
`percentile`	`vec_up`	285.0	15	8.06	\$7.08 (\$8.60)
`percentile`	`hard_tr`	212.3	15	**2.10**	\$7.05 (\$8.45)
`reciprocal`	`hard_tr`	26.9	1	0.10	\$8.95 (\$16.40)

- **Cancellation Equivalence**: Under `vector_upscale`, `target_clip + vec_up` and `target_clip + hard_tr` converge to the exact same optimal profile (ESS $= 296.4/300$, 97.0% full weight, Mean \$7.13, SD \$8.92), while `linear + vec_up` matches on all 299 active units (ESS $= 296.2$).
- **Percentile Tail Excision**: Zeroing the top 5% (15 units) in Stage 1 followed by `vec_up` boosts surviving inliers to $\alpha_i^* = 1$ (ESS $= 285.0, \epsilon_{\mathbf{y}} = 8.06$), whereas `hard_tr` yields the tightest privacy ($\epsilon_{\mathbf{y}} = 2.10$).
- **Remark 1 Validated**: Reciprocal $1/f_i$ normalization collapses effective sample size (ESS $= 26.9/300$), inflating synthetic variance (SD \$16.40).

Track 1B — Finite-Sample (ϵ, δ)-DP & Additive Nugget η



Dual-Support Plug-In (Azizi et al., Nov 2026)

- Asymp. $\epsilon_{\mathbf{x}}(0) = 2\Delta_{\alpha,\mathbf{x}} = 11.00$ leaks with finite-sample $\hat{\delta}_n(0) = 0.715$ across $R = 200$ replicates.
- `calibrate_dp_finite_sample()` freezes $B = 100$ draws $\Theta^{(1)}, \Theta^{(2)}_{\alpha}$ for **zero-retraining** η solving $\epsilon_{\mathbf{x}}(\eta) = 2(\Delta_{\alpha,\mathbf{x}} + \eta)$.

δ_{target}	$q_{1-\delta}$	**Nugget η**	**Calibrated $\epsilon_{\mathbf{x}}(\eta)$**
0.10	6.59	**+1.09**	**13.19**
0.05	6.84	**+1.34**	**13.68**
0.01	7.26	**+1.76**	**14.51**

Pipeline & Return Structures: Multi-Pass DP Synthesis (Track 1B)

```r
# 1. Calibrate privacy weights: Stage 1 target clipping + Stage 2 vector upward rescaling
dp_w <- compute_dp_weights(f_sens, stage1_method = "target_clip",
                           stage2_method = "vector_upscale", M_thresh = 5.5, Delta_goal = 5.5)
# 2. End-to-end multi-pass DP estimation (executes Pass 3 for vector upscale geometry)
dp_fit <- fit_dp_lbbnn(model_spec = list(sizes = c(p, 16, 1), problem_type = "regression"),
                       X = x_micro, y = y_sensitive, stage1_method = "target_clip",
                       stage2_method = "vector_upscale", M_thresh = 5.5, epochs = 100, lr = 0.01)
# 3. Fast dual-support plug-in finite-sample (epsilon, delta)-DP nugget calibration
dp_cal <- calibrate_dp_finite_sample(dp_fit, X = x_micro, y = y_sensitive, R = 200, delta_target = 0.05)
# 4. Generate private synthetic microdata via post-processing theorem
synth_micro <- sample_synthetic_lbbnn(dp_fit, newdata = x_micro, draws = 50)
```

Step / Object	R Class / Type	Dimensions	Statistical Contents & Role
`dp_fit$model`	`lbbnn_net` (R6)	$p \to 16 \to 1$	Final private model targeting calibrated pseudo-posterior $\xi_{\boldsymbol{\alpha}^{\text{vec}}}$.
`dp_fit$alpha`	`numeric vector`	$[n]$ records	Calibrated Stage 2 weights $\alpha_i^* = \min(1, M_{\text{thresh}}/f_i) \in [0,1]$.
`dp_fit$epsilon`	`numeric scalar`	Scalar $\epsilon_{\mathbf{y}}$	Asymptotic local DP bound $\epsilon_{\mathbf{y}} = 2\Delta_{\alpha,\mathbf{y}} \leq 2M_{\text{thresh}}$.
`dp_cal$calibration_table`	`data.frame`	3×4	Tail quantiles $q_{1-\delta}$, additive nuggets η, finite-sample $\epsilon_{\mathbf{x}}(\eta)$.
`synth_micro`	`matrix`	$[n, 50]$	50 private synthetic replicates drawn with calibrated $\hat{\sigma}$.

- **Loewner Dominance**: `"vector_upscale"` pointwise dominates hard truncation and maximizes Fisher information.
- **Post-Processing Theorem**: Downstream analysts query `synth_micro` with zero additional disclosure risk.

Section 3

PART III: Track 2 — Combining Probability & Non-Probability Samples

The Big Data Paradox in Convenience Samples (Meng, 2018)

The Non-Probability Selection Dilemma:

- Let $U = \{1, \ldots, N\}$ denote the finite target population of size N.
- In convenience/opt-in samples $S_c \subset U$, sample inclusion $I_{ci} \in \{0, 1\}$ is non-random and driven by unobserved self-selection mechanisms.
- **Meng's Law of Large Populations (2018)**:

$$\text{Bias}(\bar{y}_c) = \frac{\text{Cov}_U(I_c, Y)}{\bar{\pi}_c} = \rho_{I_c,Y} \cdot \sigma_Y \cdot \sqrt{\frac{N - n_c}{n_c}}$$

- **Exact Definitions**:
  - $\pi_{ci} = \mathbb{P}(I_{ci} = 1 \mid \mathbf{x}_i)$: Unknown propensity that unit i with covariates $\mathbf{x}_i$ self-selects into S_c.
  - $\bar{\pi}_c = \frac{1}{N} \sum_{i \in U} \pi_{ci}$: Average participation propensity across the entire target population U.
- A tiny correlation $\rho_{I_c,Y} \neq 0$ creates selection bias scaling with $\sqrt{N}$; increasing sample size n_c merely concentrates error around a severely biased estimate.

The Identification Barrier & The Conditioning Principle (Beresovsky et al., 2026)

The Missing Mass & Identification Barrier:

- Let $U = \{1, \ldots, N\}$ denote the finite target population. We observe covariates $\mathbf{x}_i \in \mathbb{R}^p$ on two samples:
  1. **Convenience Sample S_c (n_c units)**: Self-selected with *unknown* propensity $\pi_{ci} = \mathbb{P}(I_{ci} = 1 \mid \mathbf{x}_i) = \sigma(\eta_\theta(\mathbf{x}_i))$, where $\sigma(z) = \frac{1}{1+e^{-z}}$, and observed survey outcome y_i.
  2. **Reference Probability Survey S_r (n_r units)**: Selected with *known* design inclusion probability $\pi_{ri} = \mathbb{P}(I_{ri} = 1 \mid \mathbf{x}_i) > 0$, measuring shared covariates $\mathbf{x}_i$ (outcomes y_i unobserved).
- **Why Direct Binary Classification Fails**: Fitting a direct Bernoulli likelihood for $I_{ci} \in \{0, 1\}$ across U is impossible because non-sampled population units $i \in U \setminus (S_c \cup S_r)$ are completely unobserved.

The Conditioning Resolution (Quasi-Randomization vs. Pseudo-Likelihood):

- **Composite Conditioning**: Pool S_c and S_r into a combined sample S, and condition on inclusion $i \in S$. Known design probabilities π_{ri} enter $\pi_{zi} = \mathbb{P}(I_{zi} = 1 \mid \mathbf{x}_i, i \in S)$ as a structural anchor, rendering $\eta_\theta(\mathbf{x}_i)$ fully identifiable!
- **Survey Design Pseudo-Likelihood (`PSEUDO`)**: Approximate the unobserved population log-likelihood over U using Horvitz–Thompson inverse design weights $w_{ri} = \pi_{ri}^{-1}$ on S_r (Chen, Li & Wu, 2020).

Intuition & Torch Training Link for All Five Estimators

Estimator	Sample Overlap	Composite Prob π_{zi}	Torch Backprop & Loss Link
`STACK`	$S_c \oplus S_r$ (Unlinked)	$\frac{\pi_{ci}}{\pi_{ci}+\pi_{ri}}$	Offset logit$(\pi_{zi}) = \eta_\theta(\mathbf{x}_i) - \log \pi_{ri}$ in Bernoulli NLL.
`LINK1`	$S_c \cup (S_r \setminus S_c)$	$\frac{\pi_{ci}}{\pi_{ci}+\pi_{ri}-\pi_{ci}\pi_{ri}}$	Deduplicated union; backprops through $-\pi_{ci}\pi_{ri}$ overlap adjustment.
`LINK2`	$S_c \oplus S_r$ (Linked Stack)	$\frac{\pi_{ci}(1+\pi_{ri})}{\pi_{ci}+\pi_{ri}}$	Exact indicator $I_{zi} = D_i + (1-D_i)I_{ci}$; maximizes Fisher information.
`MIP`	Missing Information	Bridges `STACK` & `LINK2`	Evaluates expectation $\tilde{I}_{zi}^{\text{MIP}} = D_i + (1-D_i)\tilde{p}_i$ with $\tilde{p}_i \in [0,1]$.
`PSEUDO`	Survey-Weighted	Design pseudo-likelihood	Score: $\sum_{S_r} \pi_{ri}^{-1}$ softplus$(\eta_i) +$ $\sum_{S_c} \log(1+e^{-\eta_i}) = 0$.

- **Composite vs. Pseudo**: Composite likelihoods (`STACK`, `LINK1`, `LINK2`, `MIP`) form relative odds fractions that naturally bound loss gradients, whereas `PSEUDO` multiplies unconstrained logits by π_{ri}^{-1}.

Implementation in LBBNN (1/5): STACK (Unlinked Stacked Sample)

Statistical Formulation ($S = S_c \oplus S_r$):

- **Unlinked Multiset**: Pool S_c (n_c) and S_r (n_r) into a multiset of size $n = n_c + n_r$ with origin indicator $I_{zi} = D_i = \mathbb{I}(i \in S_c) \in \{0, 1\}$:

$$\pi_{zi}^{\text{STACK}} = \frac{\pi_{ci}}{\pi_{ci} + \pi_{ri}}, \qquad \pi_{ci} = \sigma\big(\eta_\theta(\mathbf{x}_i)\big) = \frac{1}{1 + e^{-\eta_\theta(\mathbf{x}_i)}}$$

- **Log-Odds & Rare-Event Offset** ($\pi_{ci} \ll 1$):

$$\begin{aligned}\log\left(\frac{\pi_{zi}^{\text{STACK}}}{1 - \pi_{zi}^{\text{STACK}}}\right) &= \log \pi_{ci} - \log \pi_{ri} \\ &= \eta_\theta(\mathbf{x}_i) - \underbrace{\log\left(1 + e^{\eta_\theta(\mathbf{x}_i)}\right)}_{\approx 0 \text{ when } \pi_{ci} \ll 1} - \log \pi_{ri}\end{aligned}$$

When $\pi_{ci} \ll 1$ ($n_c \ll N$), $-\log \pi_{ri}$ acts as an additive offset.

- **Exact Composite NLL**: In `nll_stack()`, we evaluate the exact fraction $\frac{\sigma(\eta_i)}{\sigma(\eta_i) + \pi_{ri}}$ without dropping $\log(1 + e^{\eta_i})$, guaranteeing exact calibration at any sampling fraction.

LibTorch Architecture (`nll_stack`):

- **Dataset Packaging**: `build_propensity_dataset(pop, method = "STACK")` binds covariates $\mathbf{X}$ with target tensor $\mathbf{Y} = [D_i, \pi_{ri}] \in \mathbb{R}^{(n_c + n_r) \times 2}$.
- **Bernoulli Composite Loss**:

$$\begin{aligned}\mathcal{L}_{\text{STACK}}(\theta) = -\sum_{i \in S_c \oplus S_r} \Big[& D_i \log \hat{\pi}_{zi}^{\text{STACK}} \\ & + (1 - D_i) \log\left(1 - \hat{\pi}_{zi}^{\text{STACK}}\right)\Big]\end{aligned}$$

- **Bounded Gradients**: Because $\pi_{ri} > 0$ sits inside the denominator $\hat{\pi}_{ci} + \pi_{ri}$, loss gradients with respect to η_i are strictly bounded in $[-1, 1]$ even when $\pi_{ri} \to 0$.

Implementation in LBBNN (2/5): LINK1 (Deterministically Linked Union)

Statistical Formulation ($S = S_c \cup (S_r \setminus S_c)$):

- **Deduplicated Union Sample**: Exact identifiers match overlapping units $S_c \cap S_r$, which are dropped from the reference block, leaving $n = n_c + n_r - |S_c \cap S_r|$ unique units.
- **Target Indicator**: $I_{zi} = I_{ci} \in \{0, 1\}$ (1 if unit $i \in S_c$, 0 if $i \in S_r \setminus S_c$).
- **Conditional Independence Overlap**: Under $\mathbb{P}(I_{ci} = 1, I_{ri} = 1 \mid \mathbf{x}_i) = \pi_{ci}\pi_{ri}$, inclusion in the union is $\pi_{ci} + \pi_{ri} - \pi_{ci}\pi_{ri}$:

$$\pi_{zi}^{\text{LINK1}} = \frac{\pi_{ci}}{\pi_{ci} + \pi_{ri} - \pi_{ci}\pi_{ri}}$$

LibTorch Architecture (`nll_link1`):

- **Dataset Packaging**: `build_propensity_dataset(pop, method = "LINK1")` filters out reference records with $I_{ci} = 1$ and constructs $\mathbf{Y} = [I_{zi}, \pi_{ri}]$.
- **Overlap-Adjusted Backprop**:

$$\mathcal{L}_{\text{LINK1}}(\theta) = -\sum_{i \in S_{\text{union}}} \Big[I_{zi} \log \pi_{zi}^{\text{LINK1}} + (1 - I_{zi}) \log(1 - \pi_{zi}^{\text{LINK1}}) \Big]$$

- Autograd differentiates directly through the non-linear product $-\hat{\pi}_{ci}\pi_{ri}$ in the denominator, correcting for the removal of duplicate units.

Implementation in LBBNN (3/5): LINK2 (Deterministically Linked Stack)

Deterministic Linking on Stacked Multiset ($S_c \oplus S_r$):

- **Multiset Retention** ($n = n_c + n_r$): Unlike `LINK1` (which drops duplicates), `LINK2` retains all stacked records—units in $S_c \cap S_r$ physically appear twice (once in S_c with $D_i = 1$, once in S_r with $D_i = 0$).
- **Deterministic Re-Labeling via Unique IDs**: Exact identifiers reveal $I_{ci} = \mathbb{I}(i \in S_c) \in \{0, 1\}$ for every unit in S_r. We define the composite indicator:

$$I_{zi}^{\text{LINK2}} = D_i + (1 - D_i) I_{ci} \in \{0, 1\}$$

  Both copies of overlapping units ($D_i = 1$ and matched $D_i = 0, I_{ci} = 1$) receive label $I_{zi} = 1$; only confirmed non-participants ($D_i = 0, I_{ci} = 0$) receive $I_{zi} = 0$.
- **Expected Instance Ratio**: Expected count of $I_{zi} = 1$ instances is $\pi_{ci} + \pi_{ri}\pi_{ci} = \pi_{ci}(1 + \pi_{ri})$ out of $\pi_{ci} + \pi_{ri}$ total instances:

$$\pi_{zi}^{\text{LINK2}} = \frac{\pi_{ci}(1 + \pi_{ri})}{\pi_{ci} + \pi_{ri}}$$

LibTorch Architecture (`nll_link2`):

- **Dataset Packaging**: `build_propensity_dataset(pop, method = "LINK2")` packs $\mathbf{Y} = [I_{zi}^{\text{LINK2}}, \pi_{ri}] \in \mathbb{R}^{(n_c+n_r)\times 2}$.
- **Loss Closure**:

$$\mathcal{L}_{\text{LINK2}}(\theta) = -\sum_{i \in S_c \oplus S_r} \Bigg[I_{zi}^{\text{LINK2}} \log\Big(\frac{\hat{\pi}_{ci}(1 + \pi_{ri})}{\hat{\pi}_{ci} + \pi_{ri}}\Big) + (1 - I_{zi}^{\text{LINK2}}) \log\Big(\frac{\pi_{ri}(1 - \hat{\pi}_{ci})}{\hat{\pi}_{ci} + \pi_{ri}}\Big) \Bigg]$$

- **Fisher Information Optimality**: Resolves label ambiguity on overlapping records in S_r, strictly maximizing Fisher information across all composite estimators.

Implementation in LBBNN (4/5): MIP (Missing Information Principle)

Probabilistic Record Linkage Origin:

- **Latent Overlap (I_{ci} Missing on S_r)**: When noisy quasi-identifiers or privacy rules preclude deterministic linking, an upstream linkage model (e.g., Fellegi–Sunter) estimates pairwise match probabilities $m_{ij} = \mathbb{P}(i \sim j \mid \mathbf{x}_i, \mathbf{x}_j)$, yielding marginal reference linkage scores:

$$\tilde{p}_i = \sum_{j \in S_c} m_{ij} \in [0,1], \qquad \tilde{\mathbf{p}} = (\tilde{p}_1, \ldots, \tilde{p}_{n_r})^\top$$

- **EM Expectation Projection (Louis, 1982)**: Replace latent I_{ci} in I_{zi}^{LINK2} with its conditional expectation $\mathbb{E}[I_{ci} \mid \text{linkage}] = \tilde{p}_i$:

$$\tilde{I}_{zi}^{\text{MIP}} = D_i + (1 - D_i)\tilde{p}_i \in [0,1]$$

- **Benchmark DGPs vs. Production**: In production, $\tilde{\mathbf{p}}$ is supplied upstream by agency linkage software. In Studies 0–3, we simulate $\tilde{p}_i = \sigma(\gamma_0 + \gamma_1 I_{ci} + e_i)$ across linkage sharpness regimes ($\gamma_1 \gg 0 \to$ `LINK2`, $\gamma_1 = 0 \to$ `STACK`).

LibTorch Architecture (`make_nll_mip`):

- **4-Column Target Tensor**: `build_propensity_dataset(..., method = "MIP", p_link = p_vec)` packs $\mathbf{Y} = [D_i, I_{ci}, \pi_{ri}, \tilde{p}_i] \in \mathbb{R}^{(n_c+n_r)\times 4}$.
- **Soft-Label Composite NLL**:

$$\mathcal{L}_{\text{MIP}}(\theta) = -\sum_{i \in S_c \oplus S_r} \Big[\tilde{I}_{zi}^{\text{MIP}} \log \pi_{zi}^{\text{LINK2}} + \left(1 - \tilde{I}_{zi}^{\text{MIP}}\right) \log(1 - \pi_{zi}^{\text{LINK2}}) \Big]$$

- **Record-by-Record Score Interpolation**:
  - High-certainty matches ($\tilde{p}_i \approx 1$) pull gradients toward the high-efficiency `LINK2` score.
  - Unmatched units ($\tilde{p}_i \approx 0$) safely revert toward the unlinked `STACK` score.

Implementation in LBBNN (5/5): PSEUDO (Design-Weighted Pseudo-NLL)

Census Approximation (Chen, Li & Wu, 2020):

- **Step 1: Unobservable Population Target (U)**: Full population census Bernoulli log-likelihood over U:

$$\ell_U(\theta) = \sum_{i \in S_c} \log\left(\frac{\pi_{ci}}{1-\pi_{ci}}\right) + \sum_{i \in U} \log(1-\pi_{ci})$$

- **Step 2: Horvitz–Thompson Substitution on S_r**: Approximate $\sum_{i \in U} \log(1-\pi_{ci})$ via inverse design weights $w_{ri} = \pi_{ri}^{-1}$:

$$\ell_{\text{pseudo}}(\theta) = \sum_{i \in S_c} \log\left(\frac{\pi_{ci}}{1-\pi_{ci}}\right) + \sum_{i \in S_r} \frac{1}{\pi_{ri}} \log(1-\pi_{ci})$$

- **Step 3: Exact Logit & Softplus NLL**: Substituting $\eta_\theta(\mathbf{x}_i) = \log \frac{\pi_{ci}}{1-\pi_{ci}}$ and $\log(1-\pi_{ci}) = -\text{softplus}(\eta_\theta(\mathbf{x}_i))$:

$$\text{NLL}_{\text{pseudo}}(\theta) = -\sum_{i \in S_c} \eta_\theta(\mathbf{x}_i) + \sum_{i \in S_r} \frac{1}{\pi_{ri}} \text{softplus}\big(\eta_\theta(\mathbf{x}_i)\big)$$

LibTorch Architecture & Low-Support Breakdown:

- **Vectorized Closure (`nll_pseudo`)**:

$$\mathcal{L}_{\text{PSEUDO}}(\theta) = \sum_{i \in S_c \oplus S_r} \Big[-D_i \eta_i + (1-D_i)\pi_{ri}^{-1} \,\text{softplus}(\eta_i) \Big]$$

- **Why PSEUDO Explodes Under Low Support**:

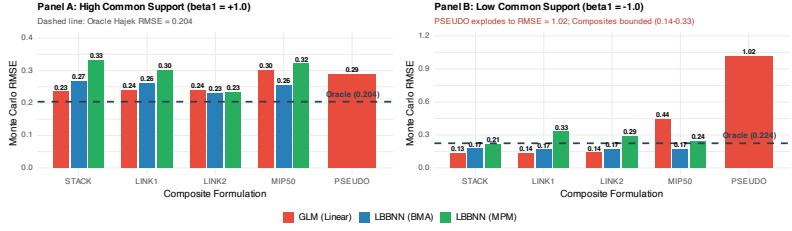
$$\frac{\partial \text{NLL}_{\text{pseudo}}}{\partial \eta_i} = -D_i + (1-D_i)\frac{\hat{\pi}_{ci}}{\pi_{ri}}$$

  - Where convenience respondents prefer strata with $\pi_{ri} \to 0$, the multiplier $\hat{\pi}_{ci}/\pi_{ri} \to \infty$.
  - Unlike composite likelihoods (gradients bounded in $[-1,1]$), unbounded gradients destabilize variational updates and collapse Kish's ESS!

Study 0 — Exact Linear Baseline Monte Carlo ($R = 20$, Beresovsky et al., 2026)

Data Generating Process ($N = 5,000, p = 3, R = 20$ Monte Carlo Replicates):

- **Covariates & Outcome**: $(X_1, X_2, X_3)^\top \sim \mathcal{MVN}(\mathbf{0}, \mathbf{\Sigma})$ ($\Sigma_{ii} = 1, \Sigma_{12} = 0.5, \Sigma_{13} = \Sigma_{23} = 0.3$); $Y_i = 1 + X_{1i} + X_{2i} + X_{3i} + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 1^2)$.
- **Sampling Designs ($n_r = 500, n_c \approx 500$)**: Reference PPS $\pi_{ri} \propto \sigma(X_{1i} + X_{2i})$; Convenience $\text{logit}(\pi_{ci}) = \beta_0 + \beta_1 X_{1i} - 0.3 X_{2i}$ (**High Support**: $\beta_1 = +1.0$; **Low Support**: $\beta_1 = -1.0$).



- **Oracle Benchmark & Stein's Phenomenon**: In Panel B (Low Support), true propensities near 0 create extreme $1/\pi_{ci}$ weights; `LBBNN` regularization smooths tail weights, achieving lower RMSE than the unregularized Oracle (0.17 vs. 0.224).
- **Pseudo-Likelihood Breakdown**: Under low support, `GLM-PSEUDO` explodes to RMSE = 1.015, while composite formulations remain bounded (RMSE ≈ 0.14–0.33).

Study 1 — Automated Confounder Selection & Noise Pruning ($p = 6$)

Data Generating Process ($N = 5,000, p = 6$):

- **Covariates**: $(X_1, X_2, X_3) \sim \mathcal{MVN}(\mathbf{0}, \boldsymbol{\Sigma})$ correlated; $X_4, X_5, X_6 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$ pure noise.
- **Non-Linear Outcome** (Y_i):

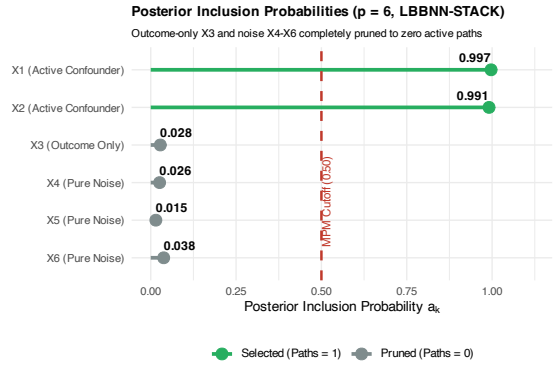
$$Y_i = 1 + X_{1i} + X_{2i} + X_{3i} + 0.5X_{1i}^2 - 0.5\sin(\pi X_{2i}) + \varepsilon_i$$

- **Reference Survey** ($S_r, n_r = 500$): Systematic PPS with $\pi_{ri} \propto \sigma(X_{1i} + X_{2i})$.
- **Convenience Sample** ($S_c, n_c \approx 500$):

$$\text{logit}(\pi_{ci}) = \beta_0 + 1.0X_{1i} - 0.5X_{2i}$$

Why Variable Selection Matters (`LBBNN-STACK`):

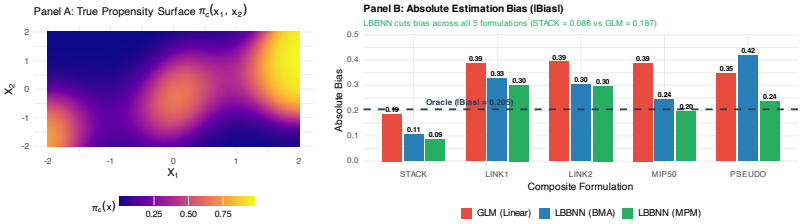
- X_3 drives Y_i but not I_{ci}; forcing X_3 into $\hat{\pi}_{ci}$ inflates Hájek weight variance without reducing bias.
- `LBBNN-STACK` (shown right; consistent across `LINK1/2/MIP`) retains confounders X_1, X_2 ($a_k > 0.99$) and prunes X_3–X_6 ($a_k \leq 0.038$, Paths $= 0$).



Study 2 — High Common Support with Non-Linear Propensity & Interaction

Non-Linear High-Support DGP ($N = 5,000, p = 6, n_r = 500, n_c = 512$):

- **Outcome & Reference PPS**: Same $p = 6$ non-linear Y_i and reference survey $\pi_{ri} \propto \sigma(X_{1i} + X_{2i})$ as Study 1.
- **True Non-Linear Convenience Propensity**:
  $\text{logit}(\pi_{ci}) = \beta_0 + \mathbf{1.0X_{1i}} - 0.5X_{2i} + 1.2X_{1i}X_{2i} + 1.0\cos(\pi X_{1i}) - 0.8X_{2i}^2$ (High overlap with π_{ri} via $+1.0X_{1i}$).

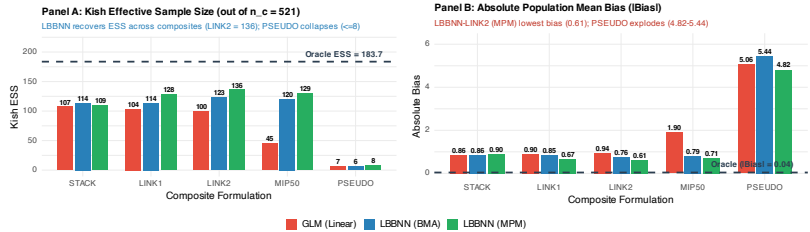


- **Functional-Form Misspecification**: Main-effects GLMs cannot represent $1.2X_{1i}X_{2i} + \cos(\pi X_{1i}) - 0.8X_{2i}^2$ ($|\text{Bias}| = 0.19$–0.39).
- **Non-Linear Recovery + Noise Pruning**: `LBBNN` learns the interaction surface while pruning X_3–X_6 ($a_k \leq 0.068$), cutting bias across all 5 formulations (`STACK`: 0.086 vs. 0.187; `LINK2`: 0.196 vs. 0.393).

Study 3 — Low Common Support + Non-Linearity: Bias vs. Kish's ESS

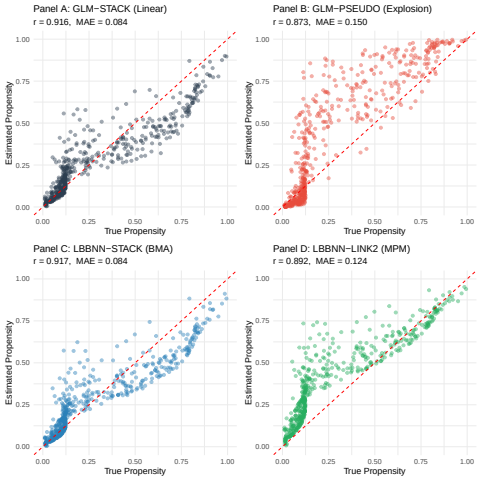
Non-Linear Low-Support DGP ($N = 5,000, p = 6, n_r = 500, n_c = 521$):

- **True Non-Linear Convenience Propensity**:
  $\text{logit}(\pi_{ci}) = \beta_0 - \mathbf{1.0X_{1i}} - 0.3X_{2i} + 0.6X_{1i}X_{2i} + 0.5\cos(\pi X_{1i}) - 0.4X_{2i}^2$.
- **Opposing Support**: Sign reversal ($-1.0X_{1i}$ vs. reference $\pi_{ri} \propto \sigma(+X_{1i} + X_{2i})$) concentrates convenience units where $\pi_{ri} \approx 0$.



- **Catastrophic Breakdown of `PSEUDO`**: Exploding π_{ri}^{-1} weights collapse Kish's ESS to $\mathbf{6.6/521}$ and explode bias $(4.82–5.44)$.
- **Composite LBBNN Stability**: `LBBNN-LINK2 (MPM)` achieves the **lowest bias (+0.61)** and **highest effective sample size (Kish ESS = 136.0)**; on `MIP50`, `LBBNN` restores ESS from 45.1 to 129.4.

Study 3 — Propensity Recovery Diagnostic (2x2 Scatterplot Grid)



Panels: (A) GLM-STACK bias from missing interactions; (B) GLM-PSEUDO severe weight explosion and scatter collapse; (C) LBBNN-STACK Bayesian Model Averaging calibration; (D) LBBNN-LINK2 Median Probability Model point recovery.

Pipeline & Return Structures: Composite Propensity (Track 2)

```r
# 1. Format combined sample with vectorized linkage probabilities p_vec
prep    <- build_propensity_dataset(pop, method = "MIP",
                                    covar_cols = paste0("X", 1:p), p_link = p_vec_r)
dl_prop <- get_dataloaders(prep$data, target_cols = prep$target_cols,
                           train_batch_size = 64, test_batch_size = 32)
# 2. Fit LBBNN with composite Bernoulli loss anchored by pi_ri offset
net_prop <- lbbnn_net(problem_type = "custom", sizes = c(p, 8, 1),
                      raw_output = TRUE, nll = make_nll_mip())
train_lbbnn(LBBNN = net_prop, train_dl = dl_prop$train_loader, epochs = 150, lr = 0.02)
# 3. Predict convenience propensities pi_c_hat & compute debiased Hajek mean
logits_c    <- predict(net_prop, newdata = prep$data[prep$Sc_indices, 1:p], mpm = TRUE)
pi_c_hat    <- as.numeric(torch::torch_sigmoid(torch::torch_tensor(logits_c)))
y_bar_hajek <- hajek_estimate(y = pop$y[pop$I.c == 1], pi_hat = pi_c_hat, dirichlet_uq = TRUE)
```

Step / Object	R Class / Type	Dimensions	Statistical Contents & Role
`prep$data`	`data.frame`	$[n_c + n_r, p + 4]$	Covariates $\mathbf{X}$ and composite target columns $[D_i, I_{ci}, \pi_{ri}, \tilde{p}_i]$.
`net_prop`	`lbbnn_net` (R6)	$p \to 8 \to 1$	Custom net with composite Bernoulli loss and structural offset π_{ri}.
`pi_c_hat`	`numeric vector`	$[n_c]$	Estimated convenience propensities $\hat{\pi}_{ci} \in (0, 1)$ on S_c.
`y_bar_hajek`	`data.frame`	Summary table	Debiased Hájek mean with 95% Dirichlet Bayesian bootstrap credible interval.

- **Vectorized Linkage**: `make_nll_mip()` ingests unit-level linkage vector $\tilde{\mathbf{p}}$ elementwise during batching.
- **Loss Stability**: Relative odds fractions naturally bound gradients, avoiding `PSEUDO` explosion.

Discussion — Bridging Track 2 Propensity Weights with Track 1 Regression

Two-Stage Quasi-Randomization (Track 2):

- **Stage 1**: `LBBNN` propensity model estimates $\hat{\pi}_{ci} = \mathbb{P}(I_{ci} = 1 \mid \mathbf{x}_i)$.
- **Stage 2**: Apply outcome-agnostic Hájek weights:

$$\hat{w}_i = \hat{\pi}_{ci}^{-1}, \qquad \hat{\bar{Y}}_{\text{Hajek}} = \frac{\sum_{i \in S_c} \hat{w}_i y_i}{\sum_{i \in S_c} \hat{w}_i}$$

  to any survey response variable y_i.
- **Outcome-Only Sparsity**: Pruning X_3 (which predicts Y but not I_c) prevents weight variance inflation.

Doubly Robust Pseudo-Posterior Imputation:

- **Bridging Track 2 → Track 1**: Normalize inverse propensity weights to sample size:

$$\tilde{w}_i = \hat{\pi}_{ci}^{-1} \frac{n_c}{\sum_{j \in S_c} \hat{\pi}_{cj}^{-1}}$$

  and feed $u_i = \tilde{w}_i$ directly into a **Track 1 LBBNN regression model** for y_i.
- **Double Robustness**: Consistent if *either* propensity model *or* outcome regression is correctly specified.
- Outcome model retains X_3 via spike-and-slab, recovering optimal predictive accuracy!

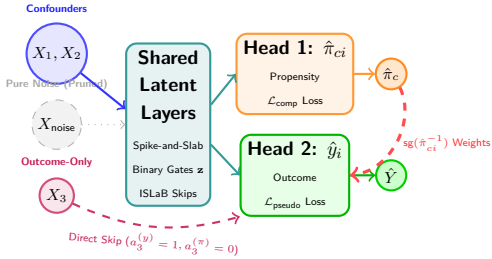
Joint Estimation — Two-Head LBBNN for Propensity & Outcome

End-to-End Joint Learning:

- **Shared Representation**: Covariates $\mathbf{x} \in \mathbb{R}^p$ pass through shared latent binary spike-and-slab layers $\mathbf{z}_{\text{shared}}$ with direct ISLaB skips.
- **Head 1 (Propensity $\hat{\pi}_{ci}$)**: Predicts $\hat{\pi}_{ci} = \sigma(\eta_\theta^{(\pi)}(\mathbf{x}_i))$ trained via composite likelihood ($\mathcal{L}_{\text{comp}}$: `MIP` or `STACK`).
- **Head 2 (Outcome $\hat{y}_i$)**: Predicts survey outcome $\hat{y}_i = \mu_\theta^{(y)}(\mathbf{x}_i, \hat{\pi}_{ci})$ using $\hat{\pi}_{ci}$ as a balancing control function, weighted by $\text{sg}(\hat{\pi}_{ci}^{-1})$.
- **Joint Variational ELBO**:

$$\mathcal{L}_{\text{joint}}(q) = \mathbb{E}_q\Big[\mathcal{L}_{\text{comp}}(\hat{\pi}_c)\Big] - \text{KL}\Big(q(\theta) \parallel p(\theta)\Big) + \sum_{i \in S_c} \mathbb{E}_q\Big[\text{sg}(\hat{\pi}_{ci}^{-1}) \log p(y_i \mid \mathbf{x}_i, \hat{\pi}_{ci})\Big]$$

- **Unified Feature Attribution**: Confounders (X_1, X_2) are shared; outcome-only drivers (X_3) feed directly into Head 2 without inflating propensity variance!



Section 4

PART IV: Software Architecture, Key Takeaways & Synthesis

Unified LBBNN Architecture Across Track 1 & Track 2

Function / Module	Operational Track	Description & Statistical Role
`get_dataloaders()`	Track 1 & 2	Handles non-trainable weights $u_i \geq 0$, normalization, missing y_i, and multi-targets.
`lbbnn_net(sigma, ...)`	Track 1 & 2	ELBO likelihood scaling by error variance σ^2 (cold-posterior protection).
`predict_domains_lbbnn()`	Track 1A (Surveys)	Vectorized domain means/totals, conditional expectation ($\hat{\mu}_i$), and Dirichlet UQ.
`impute_response_lbbnn()`	Track 1A (Surveys)	Bayesian multiple imputation for missing outcomes (y_i = NA) via posterior predictive draws.
`adjust_slab_sandwich_lbbnn()`	Track 1A (Surveys)	Williams–Savitsky (2021) Cholesky sandwich $R_2^\top R_1^{-\top}$ on active MPM slab weights.
`compute_dp_weights()`	Track 1B (Privacy)	Modular Stage 1 (`linear`, `percentile`, `target_clip`) × Stage 2 (`hard_trunc`, `vector_upscale`).
`fit_dp_lbbnn()`	Track 1B (Privacy)	Multi-pass (ϵ, δ)-PDP estimation; executes Pass 3 under vector upward rescaling.
`calibrate_dp_finite_sample()`	Track 1B (Privacy)	Dual-support plug-in finite-sample η calibration (Azizi et al., 2026; zero retraining).
`sample_synthetic_lbbnn()`	Track 1B (Privacy)	Generates (ϵ, δ)-DP synthetic microdata via the post-processing theorem.
`make_nll_mip(p_link)`	Track 2 (Propensity)	Missing Information Principle NLL supporting scalar or vectorized unit-level linkage $\bar{\mathbf{p}}$.
`nll_link1()` / `nll_link2()`	Track 2 (Propensity)	Composite Bernoulli NLL on singly matched and matched-pair samples.
`nll_stack()` / `nll_pseudo()`	Track 2 (Propensity)	Stacked pooled sample NLL and survey design-weighted pseudo-NLL.
`build_propensity_dataset()`	Track 2 (Propensity)	Formats covariates and targets for any of the 5 estimation methods.
`hajek_estimate()`	Track 1 & 2	Self-normalizing Hájek mean with optional domain matrix $\mathbf{G}$ and Dirichlet bootstrap UQ.

Key Takeaways & Synthesis for Official Statistics

1. **Survey-Weighted Pseudo-Posteriors (Track 1A)**:
   - Normalizing survey weights to sample size ($\sum \tilde{w}_i = n$) + calibrated observation noise σ^2 prevents likelihood mis-scaling ($T \neq 1$) and protects spike-and-slab KL shrinkage.
   - `LBBNN` achieves 5x lower population RMSE than classical Hájek weighting under informative sampling while pruning irrelevant covariates.
   - Frequentist UQ under P_ν: Dirichlet bootstrap restores domain CI coverage (5.5% $\rightarrow$ 96.0%), while Williams–Savitsky Cholesky sandwich restores active MPM slab coverage (18.0% $\rightarrow$ 94.0%).
2. **Multi-Pass (ϵ, δ)-PDP Microdata Synthesis (Track 1B)**:
   - Full 3×2 factorial evaluation confirms Savitsky's (2026) theory: Stage 1 direct normalization ($1 - \tilde{f}_i$, `percentile`, `target_clip`) avoids the severe ESS collapse of reciprocal $1/f_i$ weighting (Remark 1).
   - Stage 2 vector upward rescaling (`vector_upscale`) Loewner-dominates hard truncation (Prop. 2), retaining full weight ($\alpha_i^* = 1$) on safe inliers (ESS $= 296.4/300, 97.0\%$ of sample) while strictly capping sensitivity at $\Delta_{\alpha,\mathbf{y}} \leq M_{\text{thresh}}$ after Pass 3 retraining.
   - Dual-support plug-in calibration (`calibrate_dp_finite_sample`) delivers exact finite-sample $(\epsilon_{\mathbf{x}}(\eta), \delta_{\text{target}})$-DP guarantees across $R = 200$ replicates with zero neural network retraining.
3. **Composite Non-Probability Sample Integration (Track 2)**:
   - Conditioning on pooled sample inclusion $S_c \cup S_r$ allows known design probabilities π_{ri} to anchor neural network logits to the true population scale.
   - Composite Bernoulli likelihoods (`LINK2`, `LINK1`, `MIP`, `STACK`) bound loss gradients and prevent the catastrophic effective sample size collapse (`Kish ESS` < 7) of pseudo-likelihood under low common support.

Strategic Discussion — CRAN Packaging & Ecosystem Roadmap

Architectural Question: Where Should the Official Statistics & DP Modules Live?

Option A: Monolithic Integration (`LBBNN` on CRAN)

- **Single Unified Package**: Ship unit-weighted pseudo-posteriors, Williams–Savitsky sandwich UQ, finite-sample $(\epsilon_{\mathbf{x}}(\eta), \delta_{\text{target}})$-DP `fit_dp_lbbnn()`, and composite propensity losses (`STACK`, `LINK1/2`, `MIP`, `PSEUDO`) in `LBBNN` (v0.2.0).
- **Pros**:
  - Zero cross-package dependency friction; direct access to internal R6 `lbbnn_net` and LibTorch C++ closures.
  - Broad visibility to the entire `LBBNN` user base.
- **Cons**:
  - Expands the scope and vignette footprint of the core general-purpose BNN package.

Option B: Dedicated Companion (`LBBNNsurvey`)

- **Modular Extension Package**: Keep core `LBBNN` focused on foundational sparse BNNs (hooks for custom weights/losses) and release `LBBNNsurvey` importing `LBBNN`.
- **Pros**:
  - Tailored identity for official statistics agencies (BLS, Census, Statistics Norway, Eurostat).
  - Room to expand hierarchical unit-truth θ_i sandwich estimators, Fay–Herriot small area models, Fellegi–Sunter linkage bridges, and multi-stage bootstrap replicates.
- **Cons**:
  - Requires maintaining two synchronized CRAN packages.

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